

WESTCHESTER COUNTY AP, NY, USA

WMO: 725037

Lat: 41.062N

Lon: 73.705W

Elev: 367

StdP: 14.50

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94745

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.7	12.9	-7.9	3.6	11.4	-3.3	4.7	16.2	28.5	21.2	26.1	21.4	11.9	310	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	89.5	73.4	86.4	71.9	83.9	70.6	76.3	84.6	74.8	82.2	73.4	79.9	8.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	126.3	79.4	72.5	122.2	78.4	71.6	118.3	77.5	39.9	84.6	38.5	82.3	37.2	79.9	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	20.0	17.6	3.6	94.5	5.8	3.4	-0.6	96.9	-3.9	98.9	-7.2	100.8	-11.4	103.2
			1.9	78.9	5.3	1.6	-1.9	80.1	-5.0	81.0	-8.0	81.9	-11.8	83.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	29.7	32.0	39.1	49.9	59.6	68.8	74.2	72.7	65.9	54.8	44.4	35.7
	DBStd	17.19	10.15	9.21	8.71	7.79	7.12	6.25	5.02	4.90	6.28	7.49	8.06	8.60
	HDD50	2279	630	507	353	93	5	0	0	0	37	203	451	
	HDD65	5437	1093	925	803	457	202	34	2	3	65	327	617	909
	CDD50	3134	2	2	15	90	302	563	751	704	476	185	37	7
	CDD65	816	0	0	0	4	35	147	287	242	91	10	0	0
	CDH74	5874	0	0	3	61	321	1075	2256	1625	486	47	0	0
	CDH80	1687	0	0	0	15	81	298	736	453	99	5	0	0
Wind	WSAvg	7.5	9.0	9.0	9.3	8.6	7.1	6.4	5.9	5.6	6.2	7.0	8.0	8.5
Precipitation	PrecAvg	50.0	3.5	2.9	4.5	4.3	3.8	4.6	4.3	4.6	4.8	4.6	4.0	4.4
	PrecMax	61.3	7.2	7.3	7.8	11.3	7.8	8.3	8.0	9.4	13.6	16.0	6.7	6.7
	PrecMin	39.6	1.7	1.0	0.8	1.7	0.7	1.3	1.3	1.5	1.4	0.7	1.3	1.2
	PrecStd	6.3	1.4	1.3	1.8	2.0	1.6	2.0	1.8	1.8	3.1	3.4	1.5	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	61.0	71.5	81.8	87.6	90.7	94.4	91.4	87.7	79.4	67.9	62.9
		MCWB	52.5	51.7	56.7	63.2	68.9	74.0	74.9	75.5	72.2	66.2	59.5	55.5
	2%	DB	54.1	54.0	63.0	73.3	81.7	86.6	90.1	87.7	82.5	73.7	63.7	56.9
		MCWB	50.5	47.2	51.8	58.1	66.5	71.8	74.0	73.2	69.7	65.2	57.3	52.5
	5%	DB	48.3	49.3	56.8	68.0	77.3	83.5	87.0	84.5	79.3	70.2	60.7	52.7
		MCWB	44.2	43.4	48.0	54.8	63.6	70.3	72.5	71.4	68.0	62.8	55.2	49.1
	10%	DB	43.9	45.1	52.2	63.2	72.6	80.4	84.0	82.1	76.3	66.8	57.1	48.5
		MCWB	39.7	39.3	44.8	51.8	61.2	68.2	71.2	70.3	67.0	60.4	51.9	44.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	55.2	59.4	65.0	72.0	76.3	78.7	78.2	75.3	70.9	63.1	58.2
		MCDB	57.1	58.3	68.3	77.2	82.9	86.4	89.6	86.8	82.1	74.4	65.3	60.8
	2%	WB	51.1	48.7	54.2	61.1	69.1	74.3	76.6	76.1	73.2	67.5	59.6	53.6
		MCDB	53.2	52.0	59.4	69.6	78.4	83.0	85.5	83.2	78.1	72.1	62.3	55.9
	5%	WB	44.7	43.7	49.5	57.7	66.5	72.4	75.1	74.5	71.5	64.7	56.5	50.0
		MCDB	47.6	47.7	55.6	64.5	73.9	80.5	82.9	80.9	76.3	68.6	59.5	52.7
	10%	WB	40.2	39.7	45.9	54.1	63.6	70.4	73.5	73.0	69.6	61.7	53.1	44.8
		MCDB	43.6	44.7	50.8	60.3	70.0	77.2	80.1	78.8	73.9	65.8	56.5	47.6
Mean Daily Temperature Range	5% DB	MDBR	13.5	14.9	15.7	17.8	17.7	17.1	16.6	16.3	16.3	15.9	14.4	12.6
		MCDBR	18.0	20.5	22.5	25.4	24.2	20.8	19.4	18.7	19.1	19.6	18.1	17.5
		MCWBR	16.2	15.7	14.7	13.6	12.1	9.8	7.8	8.0	9.3	11.7	14.0	15.6
	5% WB	MCDBR	17.1	19.1	20.5	22.0	21.0	18.1	16.4	15.7	15.4	16.3	15.9	16.5
		MCWBR	16.7	16.6	15.9	14.1	12.4	9.8	8.0	8.0	9.6	12.2	14.6	16.5
Clear-Sky Solar Irradiance	taub		0.309	0.321	0.350	0.374	0.420	0.475	0.485	0.465	0.392	0.354	0.325	0.309
	taud		2.447	2.423	2.408	2.377	2.273	2.159	2.143	2.188	2.366	2.487	2.524	2.517
	Ebn at Noon		268	282	285	285	272	257	253	254	267	267	262	260
	Edh at Noon		25	30	33	37	42	47	48	44	35	28	23	22
All-Sky Solar Radiation	RadAvg		584	865	1183	1514	1686	1846	1890	1667	1371	939	655	491
	RadStd		46	83	95	154	168	171	98	107	125	94	57	47

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+182	+103
(47) Regional (63 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+205	+115

Nomenclature: See separate page